

A Machine-Checked Audit of the Extended-Extraction RG Contraction in the Wilson-Lattice / OS Route to a 4D Yang–Mills Mass Gap

The MeTTapedia formalization effort

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Abstract

Ben Goertzel’s manuscript *Yang–Mills Mass Gap in 4D: RG Bootstrap, Polymer Expansion, OS Reconstruction* (v14) proposes to construct a continuum $SU(N)$ Yang–Mills theory from Wilson lattice gauge theory via a multiscale renormalization-group (RG) bootstrap, a Kotecký–Preiss (KP) polymer expansion, exponential clustering, and Osterwalder–Schrader (OS) reconstruction. Its technical keystone is an *extended-extraction* contraction $\Lambda = C_1 \lambda_{\text{irr}} < 1$, with advertised constant near 2224 and $\lambda_{\text{irr}} = 2^{-13}$ at block factor 2 and extraction depth 16.

The scalar bootstrap arithmetic is exact, but the advertised Wilson recursion constant is *not derived in v14 as written*. The main-text factor product is 33152 at $b = 2$; Appendix O’s different proposed factorization is 11088/5 when its extraction factor is 2. A sign-only repair of the printed extraction series is insufficient once the displayed multivariate multiplicities are retained.

We construct a concrete finite Wilson-polynomial realization on actual group-valued link fields. In its explicit coefficient-majorant norm, canonical-dimension truncation is gauge invariant and has operator constant 1. This does not transfer from v14’s range-only specification of P_{ext} : a skew projection onto the same low-coordinate range has arbitrarily large coefficient norm. We prove the correct transfer theorem from explicit basis and dual-jet matrices with their column-sum conditioning bounds, but v14 does not provide those data for the dimension-16 action sector. A direct source reconciliation shows that they cannot be reconstructed uniquely from the present text: Appendix F.4 uses equations of motion to identify the operator space, while Appendix O.9 explicitly retains EOM operators in an off-shell basis; neither section supplies lattice Wilson representatives or a quotient basis, and Appendix I’s explicit dual jets stop at the distinct dimension-4 observable sector. Likewise, regrouping connected families by union support has norm 1; the factor 14 is exactly a *local rooted branching count* $2(2 \cdot 4 - 1)$, not a bound on the total contacts of two extended polymers. The fluctuation big- O and weighted cumulant-generation estimates also require quantitative proofs on the actual Wilson RG map.

A third machine-checked round works on the exact $SU(2)$, $b = 2$ Wilson block with normalized Haar measure and decides the extraction stage from three sides. (i) Representative-wise hard truncation is not well-defined on the actual trace-word quotient: for the centered trace words $a = 2 - \text{Tr}(U)$ and $b = 2 - \text{Tr}(U^2)$, Cayley–Hamilton forces $a^3b - 4a^4 + a^5 = 0$ with monomial canonical dimensions 16, 16, 20; truncation at 16 deletes only a^5 , and the truncated expression evaluates to -32 on the holonomy $\text{diag}(i, -i)$ while the exact expression vanishes identically. (ii) Two corrected constructions nevertheless carry exact norm-below-two extractors — the full associated-graded jet chart ($131071/65536 < 2$) and the one-plaquette trace quotient ($31/16 < 2$); with Appendix O’s other factors kept unchanged, the resulting conditional constant $90832203/40960 \approx 2217.6$ closes the two-source bootstrap at depth 16 and fails at depth 15. (iii) Neither survivor transfers to v14’s canonical (F, D) operator space: a dimension-18 scalar descendant has radial degree only 6 (resp. 3), so both extractors retain it, and the dimension-6

descendant $\partial^2 \text{Tr}(F^2)$ is zero in the manuscript’s integration-by-parts quotient yet has a nonzero gauge-invariant realization on an explicit three-plaquette block. The transfer socket required before any Wilson-RG intertwiner is therefore uninhabited as written, under both the on-shell and off-shell relation policies.

A fourth machine-checked round tests the replacement census against the exact symmetry of the finite-spacing lattice. On an explicit $\text{SU}(2)$ Cartan link chart, the sum of squared normalized Wilson potentials over the six coordinate plaquettes is gauge invariant and has canonical dimension 8. Its exact fourth jet is $6 \sum_{\mu < \nu} F_{\mu\nu}^4$. A rational orthogonal rotation changes the corresponding quartic from 1 to $337/625$. Thus the observable is hypercubic-invariant but not $O(4)$ -invariant. Every span of Appendix O.9.3’s Lorentz-scalar contractions remains $O(4)$ -invariant, so it cannot be complete for actual lattice Wilson observables. This obstruction closes the off-shell, on-shell EOM, Bianchi/IBP, trace, Cayley–Hamilton, and postponed-IBP variants of the same Lorentz-only target. It does not rule out a finite coordinate system: it forces either a genuinely hypercubic census at finite lattice spacing or a separately proved continuum/Symanzik projection with a quantitatively controlled Lorentz-breaking remainder.

Thus there is presently no derived constant C_{true} for the Wilson recursion. The support recurrence, two-marked glue identity, stopping-scale polymer estimates, Wilson reflection positivity, and continuum/OS construction remain route-specific open inputs. The finite $\mathbb{Z}_2$, $\mathbb{Z}_3$, and Q_8 gaps checked in Lean are real but Clay-irrelevant (single-link, strong-coupling, finite). The Yang–Mills mass gap remains open; we make no claim about it.

1 The problem and Ben Goertzel’s route

1.1 The Millennium problem and the lattice strategy

Fix the gauge group $G = \text{SU}(N)$ with $N \geq 2$ and let $\mathbb{R}^4$ be Euclidean space. The Yang–Mills mass-gap problem asks for the construction of a nontrivial continuum quantum Yang–Mills theory on $\mathbb{R}^4$ with gauge group G , satisfying the Wightman / Osterwalder–Schrader axioms, whose reconstructed Hamiltonian has a unique vacuum and a strictly positive spectral gap on the orthocomplement of the vacuum.

Goertzel’s v14 manuscript follows Wilson’s lattice regularization. One discretizes $\mathbb{R}^4$ as a hypercubic lattice $a\mathbb{Z}^4$ with spacing $a > 0$, with link variables $U_{x,\mu} \in G$ on oriented nearest-neighbor links. The plaquette variable is the parallel transporter around an elementary square,

$$U_{x,\mu\nu} = U_{x,\mu} U_{x+ae_\mu,\nu} U_{x+ae_\nu,\mu}^{-1} U_{x,\nu}^{-1}, \quad 1 \leq \mu < \nu \leq 4,$$

and the Wilson action penalizes deviation of plaquettes from the identity,

$$S_{\Lambda,a}(U) = \beta \sum_{p \subset \Lambda} V(U_p), \quad \beta = \frac{2N}{g^2}, \quad V(U) := 1 - \frac{1}{N} \text{Re Tr}(U) \in [0, 2].$$

Weak coupling is $\beta \gg 1$. The strategy is: (Step 1) start from the well-defined finite-dimensional lattice integral; (Step 2) run a multiscale RG bootstrap controlling the effective theory across all scales; (Step 3) establish exponential clustering at a fixed physical scale ℓ_* via a KP polymer expansion; (Step 4) transfer clustering from the coarse scale back to the fine scale with a glue / back-propagation lemma; (Step 5) reconstruct the continuum Hilbert space, vacuum, and Hamiltonian by OS, with the clustering rate becoming the spectral gap; (Step 6) verify nontriviality through a nonzero third cumulant of the action density.

1.2 The continuum-limit obstruction and extended extraction

In any RG scheme one must show that the “irrelevant” sector (operators of canonical dimension > 4) contracts at each step:

$$\|K_{j+1}^{\text{irr}}\| \leq \Lambda \|K_j^{\text{irr}}\| + (\text{source terms}), \quad \Lambda < 1.$$

The manuscript identifies a sharp obstruction in the *naive* scheme, where one extracts only the relevant and marginal operators of dimension ≤ 4 . Under rescaling, a dimension- d operator scales by b^{4-d} , so the leading irrelevant operators (dimension 6) contract by $b^{-2} = 0.25$ for $b = 2$. But the RG step also recombines polymers, contributing a combinatorial constant C_1 . The manuscript advertises the working bound $C_1 \leq 2224$ for $b = 2$; if that value were established, the naive gain would be

$$\Lambda_{\text{naive}} = 2224 b^{-2} = 2224 \times 0.25 = 556 > 1,$$

an *expanding* map: initial smallness would grow like 556^J over $J \sim |\log a_0|$ steps and diverge as $a_0 \rightarrow 0$.

The proposed resolution is *extended extraction*. Instead of extracting only dimension ≤ 4 operators, one extracts all gauge-invariant local operators up to canonical dimension $d_{\text{max}} = 16$ (for $b = 2$). The irrelevant sector then begins at dimension $d_{\text{max}} + 1 = 17$ and contracts by

$$\lambda_{\text{irr}} = b^{4-(d_{\text{max}}+1)} = b^{3-d_{\text{max}}} = 2^{-13} \approx 1.2 \times 10^{-4}.$$

With the same advertised value, the proposed contraction factor becomes

$$\Lambda = C_1 \lambda_{\text{irr}} \leq 2224 \cdot 2^{-13} \approx 0.27 < 1.$$

The manuscript presents this as genuine contraction “with substantial slack.” The audit below verifies this arithmetic only *conditional on the supplied constant*; it refutes the available derivations of that constant as written. The price is tracking a finite-dimensional vector of higher couplings $\vec{g}_j = (g_j^{(4)}, g_j^{(6)}, \dots, g_j^{(d_{\text{max}})})$ rather than the single gauge coupling; the manuscript argues these are perturbatively small and do not affect the universal one-loop beta-function coefficient $b_0 = 11/(3 \cdot 16\pi^2)$, which is determined entirely by local ultraviolet information.

1.3 The recombination constant and the extraction-projection norm

Appendix O proposes the factorization

$$C_1 \leq C_{\text{fluct}} \cdot C_{\text{cumulant}} \cdot C_{\text{recomb}} \cdot C_{\text{extract}} \leq 1.65 \times 3b^4 \times 14 \times 2,$$

which at $b = 2$ equals $11088/5 = 2217.6$, rather than exactly 2224. Appendix F.5 displays a different factor product which equals 33152 at $b = 2$; the two ledgers are inconsistent. The last Appendix-O factor $C_{\text{extract}} = \|P_{\leq d_{\text{max}}}\|_{\text{op}}$ is the operator norm of the extraction projection onto the finite-dimensional space of operators of dimension $\leq d_{\text{max}}$. The manuscript bounds it (its Computation O.7(d)) by

$$\|P_{\leq d_{\text{max}}}\|_{\text{op}} \leq \sum_{d=0}^{d_{\text{max}}} \binom{\#\text{indices}}{d} (r_1/r_0)^{-d} \leq 1 + O((r_0/r_1)^{d_{\text{max}}}),$$

and asserts $\|P_{\leq d_{\text{max}}}\|_{\text{op}} \leq 2$ for $r_1/r_0 = 1/2$ and $d_{\text{max}} = 16$. The literal negative exponent grows, but changing only its sign does not turn the displayed multivariate sum into a one-dimensional geometric series. More fundamentally, v14 specifies P_{ext} by its range and does not give the dimension-16 invariant basis, dual jets, or conditioning estimates needed to determine its operator norm. We return to the proved realization and the transfer obstruction in §3.

2 What we formalized

The formalization lives in the Lean namespace `Mettapedia.QuantumTheory.YangMills`. We summarize the four checked components: the contraction predicate, the knife-edge arithmetic, the finite strong-coupling gaps with their finite OS reconstruction, and the continuum-open gating. All theorem and definition names below are reproduced verbatim from the corresponding modules.

2.1 The extended-extraction contraction predicate

The module `Mettapedia.QuantumTheory.YangMills.RGBootstrap` models the route’s contraction obligation as a predicate over a recombination constant C , block factor b , and extraction depth $d_{\max}$. The irrelevant scaling factor is $b^{3-d_{\max}}$ (an integer power, since the useful cases have $d_{\max} > 3$), and a contraction is a nonnegative constant whose product with the irrelevant scale is below 1.

```
/-- Irrelevant scaling factor 'b^(3-dmax)' used by the extended-extraction
discussion. The exponent is an integer because the useful cases have
'dmax > 3'. -/
def irrelevantScale (b dmax : ℕ) : ℕ :=
  (b : ℕ) ^ (3 - (dmax : ℕ))

/-- The route-level contraction audit for a combinatorial constant and an
irrelevant scaling factor. -/
def HasRGContraction (C scale : ℕ) : Prop :=
  0 < C & C * scale < 1

/-- The extended-extraction audit specialized to 'scale = b^(3-dmax)'. -/
def HasExtendedExtractionContraction (C : ℕ) (b dmax : ℕ) : Prop :=
  HasRGContraction C (irrelevantScale b dmax)
```

A threshold lemma in the same module records the exact content of the predicate: for $b > 0$ and $d_{\max} \geq 3$, the contraction `HasExtendedExtractionContraction C b dmax` holds if and only if $0 \leq C < b^{d_{\max}-3}$. Thus the entire arithmetic content of the route is the comparison of C with the threshold $b^{d_{\max}-3}$.

2.2 The knife-edge arithmetic

The module machine-checks that the advertised constant contracts at $d_{\max} = 16$ and that the exact one-step gain is $2224 \cdot 2^{-13} = 139/512 < 1/3$:

```
/-- The advertised numerical slack: '2224 * 2^-13 < 1'. -/
theorem rgContraction_2224_two_sixteen :
  HasExtendedExtractionContraction 2224 2 16 := by
  refine by norm_num, by rw [irrelevantScale_two_sixteen]; norm_num, ?_
  rw [irrelevantScale_two_sixteen]
  norm_num

/-- Exact advertised one-step gain for the extended-extraction constants. -/
theorem rgGain_2224_two_sixteen_eq :
  (2224 : ℕ) * irrelevantScale 2 16 = (139 : ℕ) / 512 := by
  rw [irrelevantScale_two_sixteen]
  norm_num
```

Crucially, the same constant does *not* contract at the nearest even extraction depth below the advertised one. At $d_{\max} = 14$ the threshold is $2^{11} = 2048 < 2224$, so the contraction is arithmetically false; the module proves that any contraction at $d_{\max} = 14$ would force $C < 2048$, and hence that 2224 fails:

```

/-- Any contraction proof at 'dmax = 14' must improve the linear recombination
constant below '2048'. The manuscript's advertised bound '2224' is therefore
insufficient for this lower extraction depth. -/
theorem rgContraction_two_fourteen_forces_constant_lt_2048
  {C : } (h : HasExtendedExtractionContraction C 2 14) :
    C < 2048 := by
  have hprod : C * irrelevantScale 2 14 < 1 := h.2.2
  rw [irrelevantScale_two_fourteen] at hprod
  nlinarith

/-- Same-constant blocker at the nearest even extraction depth below '16'. -/
theorem not_rgContraction_2224_two_fourteen :
  ¬ HasExtendedExtractionContraction 2224 2 14 := by
  intro h
  have hC : (2224 : ) < 2048 :=
    rgContraction_two_fourteen_forces_constant_lt_2048 h
  norm_num at hC

```

This is the knife edge: the choice $d_{\max} = 16$ is exactly what makes the advertised constant contract, and the manuscript's own table shows $d_{\max} \in \{10, 12, 14\}$ all fail against $C_1 \leq 2224$ while $d_{\max} = 16$ succeeds. The module `Mettapedia.QuantumTheory.YangMills.RGCruX` packages this into a route-level statement: a mass-gap route that keeps the advertised constant 2224 but uses an even extraction depth strictly below 16 cannot exist, since the required contraction is arithmetically false before any OS / clustering layer is reached.

```

/-- Same-constant lower-extraction certificate in the manuscript's even
canonical-dimension shape: the cutoff is strictly below '16', is even, and uses
the same advertised recombination bound '2224'. -/
def SameConstantEvenBelowSixteenRGCertificate (dmax : ) : Prop :=
  dmax < 16 Even dmax HasExtendedExtractionContraction 2224 2 dmax

/-- No same-constant RG certificate exists for an even manuscript-shape cutoff
strictly below '16'. -/
theorem not_sameConstantEvenBelowSixteenRGCertificate
  {dmax : } :
  ¬ SameConstantEvenBelowSixteenRGCertificate dmax := by
  intro hcert
  have hdmax : dmax 14 :=
    even_lt_sixteen_le_fourteen hcert.1 hcert.2.1
  exact (not_rgContraction_2224_two_of_dmax_le_fourteen hdmax) hcert.2.2

```

A strengthened companion theorem, `not_atLeast2048EvenBelowSixteenRGCertificate`, shows that any even lower-extraction repair below $d_{\max} = 16$ would need a genuinely sharper constant ($C < 2048$), not merely the advertised one. We stress that this is an honest delimiter of the arithmetic surface, not a refutation of the route: at the raw natural number $d_{\max} = 15$ the same constant *does* contract ($2^{12} = 4096 > 2224$), and the module records this explicitly so that the obstruction is read precisely as an even canonical-dimension statement matching the manuscript's

dimensions $4, 6, \dots, d_{\max}$.

2.3 Finite strong-coupling spectral gaps and finite OS reconstruction

The module `Mettapedia.QuantumTheory.YangMills.FiniteOSReconstruction` formalizes the finite algebraic core of OS reconstruction: a reflection-positive finite kernel induces a positive pairing on observables, and a transfer operator that is self-adjoint for that pairing descends to the pairing quotient and remains self-adjoint there. On this base, the modules `Z2StrongCouplingGap` and `Z3StrongCouplingGap` build explicit finite single-link abelian lattice gauge models. For $\mathbb{Z}_2$, the one-link heat-kernel transfer at heat time 1 has eigenvalues 1 and $\exp(-1)$, so the dimensionless gap is $-\log(\exp(-1)/1) = 1$:

```
theorem z2StrongCouplingGap_eq_one :
  z2StrongCouplingGap = 1 := by
  simp [z2StrongCouplingGap, z2StrongCouplingLambdaTwo,
    z2StrongCouplingLambdaOne, z2StrongCouplingRatio, Real.log_exp]

/-- End-to-end finite lattice theorem: the explicit 'Z' strong-coupling model is
nontrivial, reflection positive, has an OS transfer operator, and has
dimensionless transfer gap at least '1'. -/
theorem z2StrongCoupling_latticeYangMills_massGap_landmark :
  _ : Z2StrongCouplingGapCanary,
    ( x y : Z2OneLinkConfig, x y)
      OSReflectionPositive z2StrongCouplingKernel
      z2StrongCouplingGap = 1
      (1 : ) z2StrongCouplingGap
      0 < z2StrongCouplingGap := by
exact
  z2StrongCouplingPositiveCanary,
  z2OneLinkConfig_nontrivial,
  z2StrongCouplingKernel_reflectionPositive,
  z2StrongCouplingGap_eq_one,
  z2StrongCouplingGap_lower_bound_one,
  z2StrongCouplingGap_positive
```

The $\mathbb{Z}_3$ module proves the analogous landmark with two mean-zero transfer eigenvectors and the same unit gap. These theorems are unconditional and non-vacuous: they exhibit genuine reflection positivity, a genuine finite OS transfer reconstruction, and a strictly positive spectral gap. They are also kept rigorously separate from any continuum claim. Each module carries an explicit scaling diagnostic showing that a finite gap need not survive a naive continuum schedule. For the heat-time schedule $t(a) = a^2$, the *physical* gap (dimensionless gap divided by lattice spacing) closes as $a \rightarrow 0$:

```
/-- A concrete gap-closing schedule for the finite model under the heat-time
scaling 't(a)=a^2': the physical gap can be made smaller than any positive
threshold by taking the lattice spacing small enough. -/
theorem z2QuadraticHeatTime_physicalGap_closes :
  : , 0 <
  a : ,
    0 < a a <
      z2HeatTimePhysicalGap a (a ^ 2) < := by
intro h
refine / 2, by linarith, by linarith, ?_
```

```

have hne : / 2 0 := by linarith
rw [z2QuadraticHeatTime_physicalGap_eq hne]
linarith

```

2.4 The continuum endpoint is gated open

The module `Mettapedia.QuantumTheory.YangMills.ProofState` records a status ledger over the route’s nodes. The finite RG / extraction surfaces are marked `checked` or `refuted` as appropriate, but the continuum mass-gap endpoint carries an explicit `openGoal` status, gated behind a constructive-QFT promotion barrier (continuum measure, Euclidean covariance, reflection positivity, OS reconstruction, and Hamiltonian mass-gap transfer — none of which is represented for the RG / extraction route, which supplies only a finite ledger):

```

/-- The final mass-gap endpoint remains an open goal in this lane. -/
def yangMillsMassGapEndpointNode : YangMillsProofNode where
  id := "yang-mills.mass-gap-endpoint"
  status := .openGoal
  truthValue := 0, 95
  evidence := "currentYangMillsMassGapEndpoint_blockedByConstructiveGate blocks promotion
    from the audited RG/extraction surfaces to a continuum mass-gap endpoint ..."
  nextObligation := "Connect the RG and extraction surfaces to a genuine continuum mass-
    gap statement only after the constructive-QFT gate, OS/Hamiltonian transfer
    dependency packet, challenge-world interface packet, and mass-gap promotion gate
    clear."

theorem yangMillsMassGapEndpointNode_open :
  yangMillsMassGapEndpointNode.status = .openGoal := by
  rfl

```

A companion theorem `currentYangMillsMassGapEndpoint_blockedByConstructiveGate` records that the endpoint is open, the constructive checks are all unrepresented, and the only supplied interface is the finite RG / extraction ledger. In short, the formalization does not over-claim: nothing in it asserts a continuum mass gap.

3 Honest status and scope

This lane is, by construction, conservative; we now state precisely what is and is not established.

The contraction constant is an input, not a conclusion. Every arithmetic theorem above takes the recombination constant as a *given real number*. The predicate `HasExtendedExtractionContraction C b dmax` is a statement about C , b , $d_{\max}$ alone; the checked facts `rgContraction_2224_two_sixteen` and `not_rgContraction_2224_two_fourteen` verify the comparison $C \leq b^{d_{\max}-3}$ for the specific value $C = 2224$. Nothing in the formalization derives that 2224 (or any $O(1) \cdot b^4$ bound) is the correct value of the manuscript’s recombination constant. The Lean ledger is therefore faithful to the route’s bookkeeping but agnostic about the route’s hardest analytic claim.

The extraction projection: realization succeeds, range-only transfer fails. The sign-only repair is insufficient: even with positive exponent, the displayed binomial/multivariate multiplicities

make the corrected finite sum exceed 2 once at least two coordinates are present. We therefore constructed the natural finite polynomial realization rather than treating the omission as a verdict. For a finite Wilson block of actual group-valued links, plaquette holonomies, and a conjugation-invariant Wilson potential, finite polynomials in the plaquette coordinates are gauge invariant. In the explicitly defined weighted coefficient-majorant norm, canonical-dimension truncation has operator constant 1 (module `WilsonBlockMajorant`).

This is our realization of the manuscript’s extraction idea, not an attribution to v14. It cannot be transferred to v14’s P_{ext} from the specified range alone: `projectionRangeSpecification_does_not_imply_bound_two` constructs an idempotent skew projection onto the same low-coordinate range that violates the bound 2. The positive transfer theorem `coefficientL1_basisJetProjection_1e` shows exactly what would suffice: explicit basis and dual-jet matrices whose coefficient-column bounds multiply. The construction attempt is certified in `Dimension16WilsonExtractionCertificate`: F.4’s on-shell relation policy and O.9’s off-shell policy disagree already on the displayed dimension-6 divergence-square operator; the exact block quotient, lattice representatives, finite basis, dimension-16 dual jets, biorthogonality, and conditioning matrices are not supplied. Appendix I’s explicit jets concern the separate observable projection $P_{\leq 4}^{\text{obs}}$. Hence the extraction component is not a standing erratum repair, and C_1 is not derived.

Hard truncation does not descend to the actual trace-word quotient. The third round replaces schematic carriers by the exact object: the $\text{SU}(2)$, $b = 2$ Wilson block with normalized Haar measure — 81 vertices, 216 positive links of which 8 are internal, 216 plaquettes, an internal spanning tree with 7 co-tree links, exact boundary-plus-co-tree coordinates, the gauge action, the normalized Wilson potential, and the fixed-boundary product-Haar reference integral with its logarithmic effective block action (modules `SU2WilsonBlockPilot`, `SU2MajorantRepairDecision`). On this object the question “is canonical-dimension truncation even well-defined on the gauge-invariant quotient?” has a negative, intrinsic answer. For any $U \in \text{SU}(2)$ the Cayley–Hamilton identity gives $\text{Tr}(U^2) = \text{Tr}(U)^2 - 2$; writing $a = 2 - \text{Tr}(U)$ and $b = 2 - \text{Tr}(U^2)$ (each of canonical dimension 4), the exact relation $b = 4a - a^2$ yields

$$a^3b - 4a^4 + a^5 = 0,$$

whose three monomials have construction-derived canonical dimensions 16, 16, 20. Truncation through dimension 16 deletes only a^5 ; on the explicit holonomy $\text{diag}(i, -i)$ the truncated expression evaluates to -32 , while the exact left-hand side vanishes on every field. Hence the semantic kernel is not stable under representative-wise hard truncation, no function on the evaluated trace-word quotient agrees with that truncation, and the corresponding descended extraction certificate is uninhabited (module `SU2CayleyHamiltonTruncationObstruction`). The witness uses only the Cayley–Hamilton relation — no equations-of-motion quotient and no boundary-extension convention — so it applies to both of v14’s relation policies, and it refutes any uniform-in- N architecture that requires representative-wise hard truncation of redundant Wilson trace words.

Corrected constructions do carry norm-below-two extractors. Truncation is, of course, definable on a *chosen basis*, and the third round constructs the two natural repairs exactly. The full associated-graded jet chart on the gauge-fixed block (7 co-tree links, 3 Lie coordinates each, hence 21 coordinates) carries the symmetric-monomial basis of jets through total degree 16, with identity jet matrix and biorthogonal coefficient duals; radial Taylor extraction at radius ratio $1/2$ has the exact operator constant $\sum_{d=0}^{16} 2^{-d} = 131071/65536 < 2$ (modules

`SU2AssociatedGradedExtraction`, `SU2AssociatedGradedExtractionDecision`). Independently, the evaluated one-plaquette trace algebra is exactly a one-variable polynomial algebra: the normal-form theorem $b = 4a - a^2$ (proved semantically complete via an infinite Cayley-rotation family) gives the basis $1, a, a^2, a^3, a^4$ through dimension 16, with extraction constant $\sum_{d=0}^4 2^{-d} = 31/16 < 2$ (module `SU2IndependentTraceBasis`). Keeping Appendix O’s other factors unchanged, the associated-graded constant yields the conditional ledger value

$$C_{\text{proposed}} = \frac{90832203}{40960} \approx 2217.583 < 2224,$$

whose two-source gain at depth 16 is $90832203/335544320 \approx 0.2707 < 1/3$, while depth 15 does not close. This is conditional bookkeeping: the fluctuation, cumulant, and recombination factors remain unproved on the actual map. Two exact theorems keep these constructions honestly separated from v14’s object: the full graded chart has dimension 12,875,774,670 and admits no linear equivalence onto any coordinate space of dimension at most 1851, so the manuscript’s schematic count requires a genuine invariant quotient that v14 does not construct; and the radial norm used here is a precise repair, not provably the manuscript’s intended multivariate norm.

Neither survivor transfers to the canonical (F, D) space. The transfer question is then decided negatively by two independent machine-checked obstructions (module `V14FDQuotientTransferNoGo`, headline `v14_fullFD_quotientTransfer_noGo`). *Derivative blindness:* v14 counts canonical dimension as $2 \cdot \#F + \#D$ (its O.9.1), but radial degree on either chart is blind to derivative factors. The scalar descendant $(\partial^2)^3[(\text{Tr} F^2)^3]$ has canonical dimension 18 yet radial degree 6 on the associated-graded chart and 3 on the trace chart, so both norm-below-two extractors *fix* it — and an explicit two-cell zero-boundary sixth-difference witness evaluates to -6 , so the retained operator is nonzero. Both survivors therefore fail to implement the dimension- ≤ 16 projection of F.4.2, independently of the choice of quotient basis (`not_associatedRadialImplementsCanonicalCutoff`, `not_independentRadialImplementsCanonicalCutoff`). *Integration-by-parts/block incompatibility:* v14’s local norm (F.3) lives on ordinary functions on a finite block retaining boundary data, while both relation branches identify integrated total derivatives with zero. The dimension-6 descendant $\partial^2 \text{Tr}(F^2)$ is gauge invariant and evaluates to 1 on an explicit three-plaquette block field (a purely quadratic associated-grade witness gives the same value, so higher Taylor terms are not the cause). An ordinary block-functional realization would have to send one quotient class to both the zero function and a nonzero boundary descendant; the faithful transfer sockets — exact-Wilson, pure associated-graded, and every RG-compatible extension — are therefore uninhabited under both relation policies. Consequently no corrected recursion constant propagates from the constructions above into the bootstrap. The obstruction is structural rather than informational: supplying the omitted dimension-16 coordinates or conditioning matrices cannot make either extractor into the canonical (F, D) projection. A viable replacement needs a derivative-aware filtration whose truncation provably agrees with $2 \cdot \#F + \#D$, together with a boundary-aware complex that retains total-derivative descendants on each block as boundary cochains and imposes integration by parts only at global recombination; constructing exactly that replacement is the current object of this formalization.

The actual-lattice census cannot be Lorentz-only. The derivative-aware boundary-cochain repair makes the filtration and local boundary bookkeeping explicit, but a further obstruction appears before its coordinate basis or dual jets can be supplied. Appendix O.9.3 enumerates only

Lorentz-invariant complete contractions. The exact finite-spacing Wilson lattice has only hypercubic symmetry, and the module `SU2LatticeFDCensusNoGo` supplies a gauge-invariant canonical-dimension 8 observable whose fourth Cartan jet is

$$6(F_{01}^4 + F_{02}^4 + F_{03}^4 + F_{12}^4 + F_{13}^4 + F_{23}^4).$$

An explicit rational orthogonal rotation changes its normalized quartic value from 1 to 337/625. Hence no finite span of the $O(4)$ -invariant O.9.3 jets contains it. The necessary spanning core, the O.9.2 off-shell certificate, and the F.4.3 on-shell certificate are all formally uninhabited. The witness is derivative-free, so changing EOM, Bianchi, integration-by-parts, trace, Cayley–Hamilton, or boundary-cochain reduction policies cannot repair this same-target symmetry mismatch. The result is already an $N = 2$ counterexample to an architecture asserted uniformly for $SU(N)$. It is not a no-go for a hypercubic-complete basis or for a controlled continuum projection.

The finite gaps are real but Clay-irrelevant. The $\mathbb{Z}_2$ and $\mathbb{Z}_3$ spectral gaps are honest theorems, but they are abelian, single-link, strong-coupling, and finite. They exemplify the finite OS-reconstruction mechanism; they are not lattice approximations to a continuum $SU(N)$ theory, and the accompanying scaling diagnostics show that the physical gap can close under a naive continuum schedule. They establish the machinery, not the target.

The continuum mass gap is correctly open. The continuum endpoint is gated `openGoal` with no claimed support, and the constructive-QFT prerequisites are all marked unrepresented. The formalization makes no claim, in either direction, about the existence of a continuum Yang–Mills mass gap.

4 What remains, exactly

Machine-checked since the first version of this report

A second formalization round closed several items that the first version listed as open, and sharpened others into theorems:

- **The two-source bootstrap threshold.** Linear contraction ($\Lambda < 1$) is *not* sufficient for the manuscript’s bootstrap; the scalar induction closes iff $a = C b^{3-d_{\max}} \leq 1/3$ (`scalar_bootstrap_two_sources`). At $C = 2224$, $b = 2$: depth 14 fails even linear contraction, depth 15 contracts linearly but the bootstrap *fails* (`not_twoSourceBootstrapSlack_2224_two_fifteen`), depth 16 closes. The true reason for $d_{\max} = 16$ is the $1/3$ threshold, not parity of the depth — and the threshold is *dynamical*: an explicit nonlinear toy orbit realizes the recursion with equality and exits the ε^* -ball in one step at depth 15 (`toyRG_ball_escape_two_fifteen`, `benToyRGBootstrapRealization`).
- **The extraction realization and transfer obstruction.** The manuscript’s literal displayed sum is ≥ 3 for every $n \geq 1$ and never ≤ 2 (`three_le_literalExtractionOperatorSum`, `not_literalExtractionOperatorSum_le_two`). On an explicit finite Wilson-block polynomial algebra, coefficient-majorant truncation is norm one and gauge invariant (`WilsonBlockMajorant`). A matrix column-sum theorem transfers through explicit basis and dual jets; v14 does not provide these for $P_{\text{ext}} \leq 16$, and a skew-projection counterexample proves that no bound follows from the range alone (`ExtendedExtractionTransfer`). The source-certificate audit further proves that F.4 and O.9 select different EOM relation policies

and records the absent lattice representatives, quotient basis, dimension-16 jets, biorthogonality, and conditioning rows (`Dimension16WilsonExtractionCertificate`).

- **The support-growth dichotomy.** The two collar-growth statements the manuscript makes are inequivalent, and both branches are now tight theorems: a coarsen-then-thicken recurrence $r_{j+1} \leq r_j/b + A$ yields the uniform bound $r_0/b^J + Ab/(b-1)$, while a thicken-only recurrence $r_{j+1} \leq r_j + A$ yields exactly $r_0 + AJ$ (`support_growth_dichotomy`).
- **Gap $\Leftrightarrow$ clustering, finite class.** Reflection positivity, self-adjointness, and a q -contraction on the vacuum complement give exponential decay of truncated two-point functions at rate $-\log q$ (`abs_truncatedTwoPoint_le_exp`); conversely, RP plus per-vector moment decay $\langle v, T^n v \rangle \leq C_v q^n$ on a dense subset gives the gap (`eigenvalue_abs_le_of_moment_decay`) — note the *per-vector* constants: only the rate must be uniform, which weakens what the manuscript’s Assumptions K.7/M.5 need to supply. A regression records that an additive t -independent tail defeats the conclusion (`moment_decay_additive_defeat`), so large-field remainders must decay, not merely be small.
- **A non-abelian instance.** The quaternion group $Q_8 \subset SU(2)$ single-link model: machine-checked non-commutativity ($ij = k \neq -k = ji$), reflection positivity, spectral gap carried by the four-dimensional spin isotypic component, and clustering; together with the $\mathbb{Z}_2$ instance this discharges the lattice-side inputs (`massGap`, `reflectionPositive`, `clustering`) of the finite continuum-gate conditional (`z2_continuumMassGap_of_reconstructionMachine`).

Machine-checked in the third round

A third round moved from schematic carriers to the exact $SU(2)$, $b = 2$ Wilson block (§3) and decided the extraction stage on it:

- **Hard truncation refuted intrinsically.** The Cayley–Hamilton relation $a^3b - 4a^4 + a^5 = 0$, with monomial dimensions $(16, 16, 20)$ and the explicit -32 evaluation, proves representative-wise canonical truncation is not well-defined on the evaluated trace-word quotient (`SU2CayleyHamiltonTruncationObstruction`).
- **Exact norm-below-two extractors constructed.** The 21-coordinate associated-graded jet chart has extraction constant $131071/65536$ and the one-plaquette trace quotient has $31/16$, both certified with explicit bases, identity jet matrices, and biorthogonal duals; the conditional ledger value $90832203/40960 \approx 2217.6$ closes the two-source bootstrap at depth 16 and fails at depth 15 (`SU2AssociatedGradedExtraction`, `SU2IndependentTraceBasis`).
- **A certified dimension separation.** The full graded chart has dimension $12,875,774,670$ and admits no linear equivalence onto any space of dimension ≤ 1851 , so v14’s schematic operator count requires an invariant quotient the manuscript does not construct (`SU2AssociatedGradedExtractionDecision`).
- **The (F, D) transfer socket is uninhabited as written.** The dimension-18/radial-degree-6 descendant and the integration-by-parts/block witness $\partial^2 \text{Tr}(F^2)$ jointly refute the faithful transfer sockets for both relation policies and all RG-compatible extensions (`v14_fullFD_quotientTransfer_noGo`).

Machine-checked in the fourth round

A fourth round constructed the derivative-aware boundary-cochain scaffold and then decided whether its dimension-16 coordinate socket could be filled by the manuscript’s Lorentz-only operator census:

- **The repaired filtration and boundary bookkeeping are mechanical.** The canonical-dimension filtration, boundary-aware cochain coordinates, coefficient projection, and conditional scalar bootstrap are machine-checked. Their analytic Wilson-RG factors remain explicit inputs.
- **The actual-lattice O.9 census is refuted at dimension 8.** The exact $SU(2)$ Wilson observable, Cartan chart, fourth-jet calculation, and rational rotation witness prove that the hypercubic quartic jet is not Lorentz invariant (`SU2LatticeFDCensusNoGo`).
- **Both relation branches and six same-target variants are closed.** Neither the off-shell O.9.2 policy nor the on-shell F.4.3 policy admits a complete Lorentz-only certificate. EOM, Bianchi/IBP, cyclic/multitrace, $SU(2)$ Cayley–Hamilton, and postponed-IBP variants share the same uninhabited spanning core.
- **The live fork is explicit.** One must either enumerate an $H(4)$ -complete finite-spacing basis and rederive its duals and constants, or prove a Symanzik/continuum projection with quantitative decay of the Lorentz-breaking artifact coordinates. No numerical factor transfers automatically from the refuted census.

The remaining inputs

With the arithmetic and structural skeleton closed, the route now rests on the following inputs — listed with their exact status in v14, since these are three different situations.

1. **Derive the extraction/recombination constant on the actual map.** *Status: not derived.* Appendix F.5’s displayed factors equal 33152 at $b = 2$; Appendix O’s different factors with $C_{\text{extract}} = 2$ equal 11088/5. The fluctuation big- O constant, weighted cumulant-generation estimate, rooted tree-graph factor, and basis/dual-jet conditioning are not proved on the actual map. Regrouping connected families by union has norm one; the factor 14 is the local rooted branching count, not a total-contact bound. The third round sharpened what a derivation must overcome: representative-wise truncation is refuted intrinsically, exact norm-below-two extractors exist on the corrected charts, and the transfer of any such extractor to the canonical (F, D) space is refuted by the derivative-blindness and integration-by-parts/block obstructions. *What settles it:* construct a derivative-aware filtration whose truncation provably agrees with canonical dimension $2 \cdot \#F + \#D$, together with a boundary-aware complex retaining total-derivative descendants per block and imposing integration by parts only at global recombination. The fourth round constructed that mechanical scaffold but proved that its actual-lattice coordinate census cannot use only O.9.3’s Lorentz scalars. *What settles it now:* enumerate and certify an $H(4)$ -complete basis through dimension 16, with exact duals and conditioning, or supply a quantitative continuum/Symanzik projection controlling every discarded Lorentz-breaking coordinate; then prove the norm-below-two estimate, the weighted Wilson cumulant/tree estimate, and the Wilson-RG intertwining law on the same object. The resulting actual constant can then be checked against $C b^{3-d_{\max}} \leq 1/3$.

2. **Which support recurrence the glue satisfies.** *Status: v14 asserts both branches of a now-proven dichotomy in different places. What settles it:* one paragraph identifying the recurrence. If coarsen-then-thicken, the uniform bound is already a theorem; if thicken-only, the $O(J)$ growth breaks uniform tree decay and the step needs a new argument.
3. **The two-marked correlation identity.** *Status: missing.* The glue / back-propagation step requires a *two*-marked transport identity of the shape $\langle A; \tau B \rangle_{\mu_0} = \langle QA; \tau' QB \rangle_{\mu_J}$; v14 states the one-marked pushforward duality, and the two-marked version does not follow from it (conditional expectations do not factor). *What settles it:* state it precisely and prove it, or carry it as an explicit named assumption; once stated, its formalization is immediate.
4. **KP convergence at the stopping scale and Wilson-measure reflection positivity.** *Status: abstract results are classical; the route-specific instantiation is open.* One must construct the actual stopping-scale Wilson activities in the same norm, prove the KP criterion uniformly, and verify reflection and boundary-limit compatibility for the same Wilson family and RG kernels.
5. **The real RG map satisfies the recursion.** *Status: the central claimed content of v14's RG analysis; not independently verified here.* Everything above shows the surrounding machine is consistent and sharp; whether the actual Yang–Mills RG map, with its actual constants, satisfies the two-source recursion inside the $1/3$ -threshold is the mathematical core of the route — the one input that is neither bookkeeping, nor a clarification, nor classical.

Items (1), (2), (3), and (5) contain route-specific mathematics. The classical theorem statements in item (4) do not discharge their Wilson instantiation. The actual RG recursion and constructive-QFT promotion remain open, and the Yang–Mills mass gap remains open.

Lean modules. The results reported here are in `Mettapedia.QuantumTheory.YangMills.RGBootstrap` (the contraction predicate and knife-edge arithmetic), `Mettapedia.QuantumTheory.YangMills.RGCruX` (the even lower-extraction route obstruction), `Mettapedia.QuantumTheory.YangMills.FiniteOSReconstruction`, `Mettapedia.QuantumTheory.YangMills.Z2StrongCouplingGap`, and `Mettapedia.QuantumTheory.YangMills.Z3StrongCouplingGap` (the finite OS reconstruction and strong-coupling gaps), and `Mettapedia.QuantumTheory.YangMills.ProofState` (the continuum-open gating); and, from the second round, `RGBootstrapSlack` (two-source bootstrap and depth trichotomy), `ExtractionMajorant` (displayed-sum audit and generic majorant model), `WilsonBlockMajorant` (the concrete finite Wilson-polynomial realization), `ExtendedExtractionTransfer` (the range-only obstruction and explicit basis/dual-jet transfer theorem), `Dimension16WilsonExtractionCertificate` (the F.4/O.9 relation-policy mismatch, checked source transfer table, and obstructed as-written coordinate certificate), `WilsonPolymerRecombination` (norm-one union regrouping, the total-contact counterexample, and the rooted interpretation of 14), `WilsonExtractionRecombinationConstant` (the two ledger calculations, analytic-factor obstructions, and conditional depth arithmetic), `SupportGrowth` (the recurrence dichotomy), `TransferGapClustering` and `Z2TransferClustering` (gap $\Leftrightarrow$ clustering and the discharged finite continuum gate), `ToyRGMap` (the dynamical realization and depth-15 escape), and `Q8StrongCouplingGap` / `Q8TransferClustering` (the non-abelian instance); and, from the third round, `SU2WilsonBlockPilot` (the exact $SU(2)$ block, tree gauge, co-tree coordinates, and fixed-boundary Haar reference integral), `SU2CayleyHamiltonTruncationObstruction` (the intrinsic non-descent witness), `SU2MajorantRepairDecision` (the sealed truncation-repair decision), `SU2AssociatedGradedExtraction` / `SU2AssociatedGradedExtractionDecision`

(the graded jet basis, its exact extraction constant, and the dimension separation from the schematic count), `SU2IndependentTraceBasis` (the one-plaquette normal form and its extraction constant), and `V14FDQuotientTransferNoGo` (the derivative-blindness and integration-by-parts/block obstructions and the uninhabited transfer sockets); and, from the fourth round, `V14BoundaryCochainComplex`, `V14BoundaryCochainExtraction`, and `V14BoundaryCochainBootstrap` (the derivative-aware boundary-cochain scaffold and its conditional arithmetic), together with `SU2LatticeFDCensusNoGo` (the exact hypercubic quartic witness and the off-shell/on-shell Lorentz-census no-go) — all in the same namespace, all axiom-guarded to `{propext, Classical.choice, Quot.sound}` with no `sorry`. The primary mathematical source is Ben Goertzel’s manuscript *Yang–Mills Mass Gap in 4D: RG Bootstrap, Polymer Expansion, OS Reconstruction* (v14).